

CLUBBY - Comprehensive Development Plan

Product Requirements Document (PRD)

1. Introduction

1.1 Purpose

The purpose of this document is to outline the product requirements for CLUBBY, an innovative wallet application aimed at connecting retail businesses with customers. The application will allow businesses to manage customer clubs and promotions seamlessly while providing customers with a consolidated platform for all their retail benefits.

1.2 Scope

This document details the functional and non-functional requirements for both the mobile application for customers and the web/application for business owners.

1.3 Definitions and Acronyms

- **CLUBBY:** The wallet application for the retail sector.
- **User:** An individual customer using the mobile application.
- **Business Owner:** An individual or entity managing a retail business using the web/application.
- **QR Code:** A machine-readable code used for joining business clubs.
- **Promotion:** Discounts, offers, or deals provided by businesses to customers.

2. Overall Description

2.1 Product Perspective

CLUBBY aims to serve as a bridge between retail businesses and customers, providing a unified platform for managing customer clubs, promotions, and benefits.

2.2 Product Functions

- **Customer Mobile Application:** Registration, QR code scanning, benefit management, notifications, and business discovery.
- **Business Owner Application:** Dashboard, user analytics, promotion management, financial transactions, and customer interaction.

2.3 User Classes and Characteristics

- **Customers:** Individuals using the mobile application to manage their retail benefits.

- **Business Owners:** Individuals using the web/application to manage their business promotions and customer interactions.

2.4 Operating Environment

- **Mobile Application:** iOS and Android platforms.
- **Web/Application:** Web browsers (Chrome, Firefox, Safari) and mobile web access.

2.5 Design and Implementation Constraints

- Compliance with data protection regulations (GDPR).
- Secure authentication and data encryption.
- Scalable infrastructure to support a large number of users.

3. Functional Requirements

3.1 Mobile Application for Customers

3.1.1 User Registration and Profile Management

- **FR1.1:** The system shall allow users to register with their email or social media accounts.
- **FR1.2:** The system shall allow users to create and manage their profiles.

3.1.2 QR Code Scanning and Business Enrollment

- **FR2.1:** The system shall enable users to scan QR codes to join business clubs.
- **FR2.2:** The system shall automatically register users to the business club upon scanning the QR code.

3.1.3 Virtual Wallet

- **FR3.1:** The system shall consolidate all user benefits, coupons, and vouchers in a virtual wallet.
- **FR3.2:** The system shall display the expiry dates of all benefits and notify users accordingly.

3.1.4 Notifications and Alerts

- **FR4.1:** The system shall send notifications for expiring benefits.
- **FR4.2:** The system shall send alerts for new promotions and updates from businesses.

3.1.5 Business Discovery

- **FR5.1:** The system shall allow users to search for businesses by category or location.
- **FR5.2:** The system shall display detailed information about businesses, including contact details and opening hours.

3.2 Web/Application for Business Owners

3.2.1 Dashboard

- **FR6.1:** The system shall provide a dashboard displaying key metrics such as registered users, new users, and sales data.
- **FR6.2:** The system shall allow business owners to customize their dashboard view.

3.2.2 User Analytics

- **FR7.1:** The system shall provide detailed analytics on customer demographics and purchasing behavior.
- **FR7.2:** The system shall display customer locations on a map interface.

3.2.3 Promotion Management

- **FR8.1:** The system shall enable business owners to create, manage, and schedule promotions.
- **FR8.2:** The system shall provide tools for targeting specific customer segments with promotions.

3.2.4 Financial Transactions

- **FR9.1:** The system shall integrate with a secure payment gateway for handling promotion fees.
- **FR9.2:** The system shall provide detailed transaction history for business owners.

3.2.5 Customer Interaction

- **FR10.1:** The system shall allow business owners to send direct messages and notifications to customers.
- **FR10.2:** The system shall provide tools for managing customer feedback and support.

4. Non-Functional Requirements

4.1 Performance Requirements

- **NFR1:** The system shall handle up to 100,000 concurrent users without performance degradation.
- **NFR2:** The system shall respond to user actions within 2 seconds.

4.2 Security Requirements

- **NFR3:** The system shall use end-to-end encryption for data in transit and at rest.
- **NFR4:** The system shall comply with GDPR and other relevant data protection regulations.

4.3 Usability Requirements

- **NFR5:** The system shall provide a user-friendly interface for both customers and business owners.
- **NFR6:** The system shall offer multilingual support.

4.4 Reliability Requirements

- **NFR7:** The system shall have an uptime of 99.9%.
- **NFR8:** The system shall provide real-time monitoring and alerting for system issues.

CLUBBY - Comprehensive Development Plan

1. Technology Roadmap

1.1 Phase 1: Initial Development (Month 1)

- **Cross-Platform Application:**
 - Basic UI/UX design.
 - User registration and profile management.
 - QR code scanning and business enrollment.
 - Virtual wallet setup.
 - Initial notification system.
- **Backend:**
 - Basic API setup.
 - Authentication and user management.

1.2 Phase 2: Intermediate Development

- **Cross-Platform Application:**
 - Enhanced wallet functionalities (coupon management, notifications).
 - Business discovery and detailed pages.
 - Manual addition of benefits.
- **Backend:**
 - Enhanced API functionalities.
 - User analytics and promotion management.

1.3 Phase 3: Advanced Development

- **Cross-Platform Application:**
 - Enhanced notification system with geo-fencing.
 - Story feature for business promotions.
 - Integration with payment gateways for adding funds.
- **Backend:**
 - Comprehensive user interaction tools.
 - Advanced financial transactions and history.
 - Final testing and deployment preparations.

3. Information Security Strategy

3.1 Security Objectives

- Protect user data from unauthorized access.
- Ensure data integrity and confidentiality.

- Comply with relevant data protection regulations (e.g., GDPR).

3.2 Security Measures

3.2.1 Authentication and Authorization

- Use OAuth 2.0 for secure authentication.
- Implement JWT for session management.
- Role-based access control (RBAC) for different user levels.

3.2.2 Data Protection

- Encrypt data in transit using TLS/SSL.
- Encrypt data at rest using AES-256.
- Regular security audits and vulnerability assessments.

3.2.3 Application Security

- Secure coding practices (e.g., input validation, output encoding).
- Regular security patches and updates.
- Use of security libraries and frameworks (e.g., OWASP).

3.2.4 Network Security

- Use firewalls to protect backend infrastructure.
- Implement intrusion detection and prevention systems (IDS/IPS).
- Secure API endpoints with rate limiting and IP whitelisting.

3.2.5 Compliance and Privacy

- Ensure compliance with GDPR and other data protection laws.
- Regularly review and update privacy policies.
- Provide users with options to manage their data and privacy settings.

3.3 Incident Response Plan

- Establish an incident response team.
- Define procedures for detecting, reporting, and responding to security incidents.
- Conduct regular drills and reviews of the incident response plan.

This comprehensive document outlines the PRD, system architecture, technology roadmap, development timeline, and information security strategy for CLUBBY. By following this plan, we aim to deliver a secure, user-friendly platform that connects retail businesses with customers, providing significant value to both parties within a three-month development period.